

APPLICABILITY OF DECISION TREES FOR SITE SPECIFIC IRRIGATION MANAGEMENT

by

Trevor Askins

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As members of the Master's Committee, we certify that we have read the thesis prepared by Trevor Askins, titled 'Applicability of Decision Trees for Site-Specific Irrigation Management' and recommend that it be accepted as fulfilling the thesis requirement for the Master's Degree.

_____ Date: _____

Derek Groenendyk

We hereby certify that I have read this thesis prepared under my direction and recommend that it be accepted as fulfilling the Master's requirement.

HAS

Candidate Email:

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Abstract

Efficient irrigation management requires balancing adequate root-zone water availability with the prevention of excessive deep percolation, nutrient loss, and unnecessary water consumption. Currently, site specific practices rely on interpolation of previous scientific studies and lessons learned. Manually creating site-specific models to guide sensor network design would be prohibitively time consuming due to the sheer number of potential sensor combinations. This creates a gap between existing crop data and site-specific applications. This study evaluates cost-aware decision tree regression as a methodology for optimizing soil sensor placement in irrigated agricultural systems. HYDRUS-1D simulations were used to generate synthetic datasets representing pressure head, volumetric water content, and water flux for 45 representative agricultural soils cultivated with corn and beans under flood irrigation. Variable root depths throughout the growing season were incorporated to produce representative training data for machine-learning models. Decision trees were trained to predict deep percolation, average root-zone water content, and a combined monitoring objective. A modified cost-aware decision tree algorithm was then used to balance predictive accuracy against sensor installation, equipment, and measurement costs. Results indicate that moderate cost weighting frequently improved prediction accuracy while substantially reducing the number and overall cost of required sensors, suggesting that cost weighting acts as an effective regularization strategy. The optimized monitoring networks reduced deployment costs while maintaining high predictive performance, with root-zone water-content predictions achieving errors below those commonly associated with high-quality field measurements. Sensor selection varied with crop type and monitoring objective, indicating that no universally optimal sensor layout exists across agricultural settings. The proposed methodology provides a practical framework for designing cost-effective irrigation monitoring networks and demonstrates the potential of integrating numerical vadose zone modeling with machine-learning techniques to support precision irrigation.

1. Introduction

Efficiency in agriculture has long been recognized as a critical factor in maintaining global food security by maximizing production while minimizing resource use (FAO, 2023). More recently, improving production efficiency has become increasingly important for the economic viability of family-operated farms, which often face greater financial constraints and reduced economies of scale compared with large commercial agricultural operations (Lowder et al., 2021). Although large-scale agriculture plays an essential role in global food production, family farms continue to provide substantial social, cultural, and economic value within rural communities (FAO, 2014). Consequently, identifying practical methods for reducing production costs without sacrificing yield is of particular importance for smaller farming operations. One area of agriculture that has been highly studied is irrigation efficiency. Water is a major cost to farmers and reducing consumption helps the financial bottom line and environmental longevity. For the past century, papers have been published on the optimization of irrigation methods to minimize water consumption on farms (Israelsen & Hansen, 1962; Howell, 2001; Pereira et al., 2012). Yet, irrigation remains equally a science and an art. Overarching improvement to the science of irrigation is possible, but difficult due to a lack of homogeneity across agricultural sites. Site specific improvement to irrigation is more approachable as it offers less resistance.

Most scientific data on irrigation methods generalize environmental, soil, and crop conditions (Allen et al., 1998; Jones, 2004; Pereira et al., 2012). However, generalized findings require interpolation to be used to guide site-specific irrigation practices (Hillel, 1998; Šimůnek et al., 2016). When insufficient water is retained within the crop root zone, plants experience water stress that reduces growth and crop yield (Allen et al., 1998). Conversely, excessive soil moisture can reduce oxygen availability within the root zone, producing conditions that have too little plant-available oxygen and similarly reduce plant productivity (Drew, 1997; Colmer & Voesenek, 2009). Unfortunately, monitoring average water content across the entirety of a soil profile is neither practical nor optimal. Root water uptake is typically concentrated in the upper portion of the active root zone, although the exact distribution varies among crops and environmental conditions (Feddes et al., 1978; Allen et al., 1998). This challenge is further complicated by changing root depths throughout the growing season, which alters the spatial distribution of water uptake as crops mature (Feddes et al., 1978). This suggests that knowledge of both the total quantity and the vertical distribution of soil moisture is critical for irrigation management.

In addition to maintaining an acceptable range of water content, water flux through the root zone must remain sufficiently high to leach accumulated salts to prevent salinity stress (Ayers & Westcot, 1985). Excessive water flux, however, promotes nutrient leaching below the root zone and unnecessarily increases irrigation and, potentially, fertilization demands (Cameron et al., 2013). Thus, effective irrigation management requires optimizing several hydrogeologic processes rather than optimizing any single state or flux.

To improve irrigation performance at a given site, soil moisture and matric potential measurements are frequently incorporated into precision irrigation systems (Evelt et al., 2012; Kim et al., 2008; O'Shaughnessy et al., 2019). When sufficient sensor information is available, irrigation scheduling can be adjusted to maintain favorable soil-water conditions while minimizing water loss (Evelt et al., 2012). However, installing dense sensor networks is expensive due to equipment, installation, and long-term maintenance costs. Although these costs can often be justified on large commercial farms, they may be prohibitive for smaller agricultural operations. This study investigates whether cost-aware decision tree regression can identify sensor configurations that provide high predictive accuracy while minimizing installation cost. In the context of the preceding discussion, optimal monitoring considers both root zone water content, deep percolation, and changing root depths during a growing season.

The objective of this research is to evaluate whether cost-aware decision tree regression, as implemented for soil reclassification, provides a practical methodology for optimizing irrigation sensor placement (Groenendyk, 2025). By investigating multiple soils and two distinct crop plants (corn and beans), the research investigates whether it is possible to proscribe a universal sensor layout, or whether optimization must be soil- and crop-specific. Because machine-learning models require large quantities of representative training data, generating an adequate experimental dataset through field observations alone would be prohibitively expensive and time-consuming. Furthermore, there are currently no methods available to measure water flux in the vadose zone directly. To address these limitations, numerical models were used to generate both the observations and the predictions of interest. Specifically, soil, climatic, and crop characteristics were represented using HYDRUS-1D (Simunek et al., 2008; Simunek et al., 2016), which has been extensively validated for simulating variably saturated water, heat, and solute transport through variably saturated media including root water uptake. HYDRUS data points that could be represented by physical sensors (pressure and water content) were then used to train and test decision trees.

2. About Decision Trees

Decision (or regression) trees are a specific type of machine learning (Breiman et al., 1984). One key characteristic of decision trees that makes them ideally suited to this investigation is that the importance weighting of observations (referred to as features, more generally) is defined intrinsically as part of the training process. Therefore, the trained decision tree can be used directly to identify the value of information for candidate observations (Murray, 2024). This includes identifying which observations can be eliminated without affecting the performance of the trained tree. One well-recognized limitation of decision trees is that they are 'greedy' algorithms prone to 'brittleness' (Breiman et al., 1984; Hastie, Tibshirani & Friedman, 2009). As a result, they can lead to multiple solutions, which rely on different data, depending upon the random choices inherent in training. This can lead to the use of multiple different data types and sensor locations without consideration of the practical implications or requiring

different data sources. Groenendyk (2025) extended decision trees to be cost-aware by incorporating considerations of the cost of the choice of sensors lower in the tree based on the measurement types and sensor locations chosen in higher levels. With this addition, cost-aware decision trees can consider a very large, proposed data set and return optimized monitoring networks with a range of weights applied to the cost of the network.

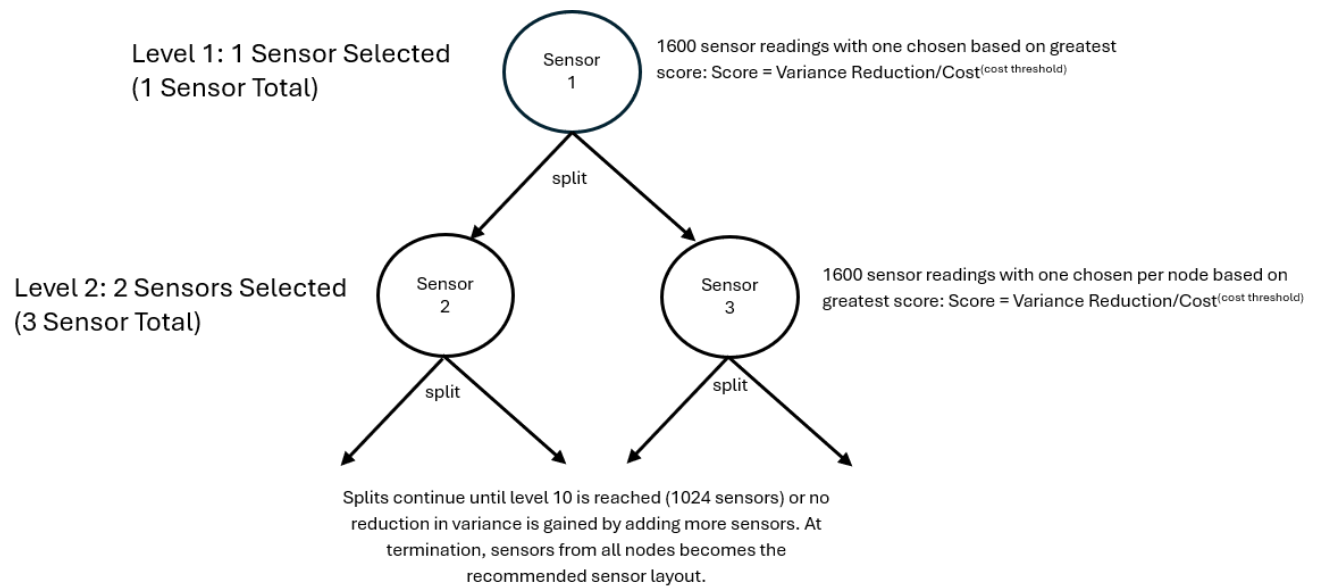


Figure 1: The method used to build the network by evaluating all options at each node and adding nodes until no improvement is possible.

It is worthwhile to discuss cost-aware decision trees in more detail because they lie at the heart of this analysis. In simple terms, a decision tree considers all available data and examines their correlation with outcomes of interest. This requires a large set of observed inputs and results for training. As discussed above, we assume that HYDRUS-1D provides a robust representation of vadose zone processes and then we develop a decision tree surrogate for this numerical model. That is, HYDRUS-1D is used to generate paired sets of water content and pressure head profiles and water fluxes through time for a wide range of soils and two key crop types. This entire set of observations (water content and pressure head profiles through time) are trained to predict key outcomes (average water content in the root zone and deep percolation).

In general, decision tree training occurs one level at a time (Figure 1) and assesses variance of the entire system (how spread out the actual answers are). At each level, the set of outcomes of interest that have been passed to that level from the next higher level are considered collectively. The variance of this set is determined. Then, the set is bisected based on each candidate observation and a chosen threshold value of that observation. The population-weighted sum of the variances of those two populations is determined. The observation and threshold that leads to the greatest reduction in variance is chosen. If the best observation

reduces variance of the entire tree, a new split occurs resulting in two additional nodes at the next level. Because the sensor type, placement, and time from a parent node will not be chosen in subsequent nodes, splits have fewer sample of outcomes of interest as the process continues. Recognizing both measurement error and the shared information inherent in many observation sets, the goal is not to find the single optimal split at each level. Rather, the focus of constructing decision trees is usually the performance of the tree, with minimal interest paid in the number of compositions of the observations used. This is efficient for training decision trees for prediction, but it introduces some uncertainty into the use of these codes to identify optimal observation sets. For this reason, it is advisable to repeat the analysis several times to ensure that consistent observation sets are selected. For this analysis each set of decision trees was trained and tested on four different 80/20 splits of the same data set. The results were stable, showing an R^2 range between 0.05 and 0.08 depending on the objective. Also of note, each level in a standard decision tree does not consider which observations have been used in higher levels. Again, this is efficient for decision trees that are aimed at prediction accuracy without consideration of which features are included. But, as detailed below, it fails to produce cost-effective monitoring networks because it does not consider the costs of purchasing and installing multiple sensors.

Once a decision tree is trained, it is used to determine the importance of the observations that were used. To do this, the reduction in the variance is determined at each node. This variance reduction is weighted by the number of outcomes of interest in the population that is subjected to that split. These population-weighted variance reductions are separated by observation. For this application, each distinct combination of type of measurement, depth of measurement, and measurement time constitutes a different candidate observation. Then the weighted variance reductions are summed over the entire tree for each candidate observation. Finally, these summed weighted variance reductions are normalized by their sum over the entire tree (over all observations) to define the relative importance of the candidate observation to the prediction. As a result, it can be assessed how much of the initial variance is accounted for by the entire model and how much of the total variance is explained per node. If variance explained = 0.90, it means the tree's splits (which sensors, which thresholds) account for 90% of why the trials' flux values differ from each other; only 10% of the spread is left as prediction error. If variance explained were 0, the tree would be doing no better than just guessing the average flux for every trial regardless of any sensor reading. This is powerful data that is implicit for this machine learning method. Importance can be determined for other machine learning algorithms, but it generally requires a separate analysis after training, which can be computationally expensive (Guyon & Elisseeff, 2003).

The training of a decision tree includes multiple user-selected settings. These include the basis for quality assessment (e.g. variance reduction), the required improvement to establish a new level, the minimum number of outcomes of interest required to establish a split group, and the number of levels allowed (Breiman et al., 1984; Hastie et al., 2009). In this analysis, the

trees were limited to ten splits to limit the maximum size of the sensor network to 1023 readings across the 100 available sensors, which is larger than any practical network.

Cost-consideration was implemented as an adjustable user-defined parameter. Specifically, the user chose a relative cost weighting that ranged from 0 (no cost consideration, only prediction quality) to 1 (no prediction quality consideration, only cost). Then, as for a traditional decision tree, the variance reduction was determined for each sensor and each threshold. Finally, the impact of each sensor on the total cost of the sensor layout was determined. How much a sensor reading contributed to the total cost of a network only included the cost of the physical sensor if it had not been selected for a previous layer. Similarly, the cost of installation was only included once for each sensor. Finally, every selection incurred a cost of measurement, which would include all data collection, maintenance and data communication and storage. In this way, less costly sensors could be identified while also encouraging the algorithm to reuse sensors rather than selecting new sensors or sensor depths. The variance reduction and cost were combined using:

$$\text{Score} = \text{Variance Reduction} / \text{Cost}^{(\text{cost threshold})}$$

The sensor with the highest score was selected for the layer. For example, if two sensors are compared at different cost thresholds (Table 1) sensor A is selected under low cost-weighting but sensor B is selected as cost-weighting increases (Table 2).

Sensor	Type	Depth	Variance Reduction	Marginal Cost
A	Pressure (h)	176cm	1000	\$10.00
B	Water Content (θ)	16cm	750	\$5.00

Table 1: Two theoretical sensors, their contribution to variance reduction, and the marginal cost for the sensor. These factors all get considered when the decision tree implements one sensor over the others.

	Score A Equation	Score A Raw	Score B Equation	Score B Raw
Cost Weight: 0.0	$1000/10^{0.0}$	1000	$750/5^{0.0}$	750
Cost Weight: 0.5	$1000/10^{0.5}$	316	$750/5^{0.5}$	335
Cost Weight: 1.0	$1000/10^{1.0}$	100	$750/5^{1.0}$	150

Table 2: The resulting scores of the two example sensors from Table 1 when different cost weights are applied. The sensor that would be selected under each condition is shaded.

A unique difference between the cost unaware trees and the cost weighted trees was that the cost unaware trees evaluated variance reduction infinitely, sometimes relying on 17 significant figures to determine if a sensor improved a network. For the cost weighted trees evaluation of variance reduction was limited to seven significant figures. This change effectively redefined the definition of what a “meaningful” improvement to a network was for a cost aware tree and is the reason for different sensor selections between the cost unaware tree and the 0.0 cost weighted trees in the results below.

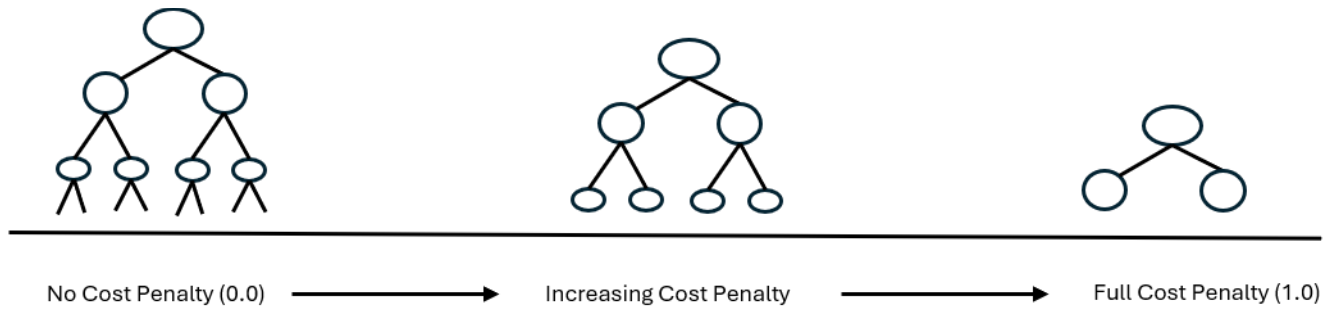


Figure 2: The expected progression of tree levels as cost thresholds increase. While the number of levels and nodes often do decrease with increasing cost threshold this is not a hard-coded rule the decision tree must follow. Under certain circumstances the number of levels and nodes can increase with increased cost threshold.

There are two additional features of the decision trees used in this analysis that are important to highlight. First, while the goal of cost-weighting was to make subsequent trees inside an objective smaller (less sensors) and cheaper (Figure 2), this was not a directive of the tree itself. All trees were blind to the results of the others, and therefore there were instances where the number of sensors selected, and the associated costs of the sensor network, went up when cost weighting increased. This is indicative of the fact that two cheaper sensors may do the work of one, but the combined cost of those two sensors is greater than the cost of the more expensive sensor. Secondly, a node identified the type of sensor, the depth of the sensor, and the time at which that sensor should be read. A sensor may be selected to be read only once during an irrigation period, or it may be selected to be read every day during an irrigation period. These temporal selections vary widely by crop and objective, do not dictate the physical layout of a sensor network, and marginally affect the total cost. Therefore, for the sake of focused discussion relevant to the research questions, the details of when readings were selected to take place will not be discussed in the main results section. Discussion on timing will be addressed in the discussion section, for consideration in future work.

3. Methods and Data

Many representative agricultural fields were defined by selecting a range of soils, crops, and irrigation parameters representative of flood-irrigated agricultural systems. Parameter values were systematically varied among simulations to generate a diverse collection of hydrologic conditions suitable for machine-learning training. Although the resulting set of conditions is hypothetical, as explained above, this methodology is necessary to provide enough observations

and outcomes for training, and it is intended to test the applicability of decision-tree-based sensor optimization for sensor network design applied to irrigation. Practical implementation would require repeating the simulation procedure using site-specific soil hydraulic properties and management practices.

It is important to note that the resulting optimization recommendations apply to all conditions used for training. For example, consider a network that is trained for all soils for corn, with flood irrigation, considering the entire growing season for a cost-weighting of 0.5. This would return the single monitoring network that applies over all these conditions. If someone wanted to examine, for example, a network that is best suited to a narrower range of soils, then the analysis would have to be re-run using only those soils in the training. Similarly, if someone was interested in a single network that produced the optimal monitoring for both corn and beans, then they would have to retrain the network including results for both crops. For this study all the analyses were conducted for the full range of soils considered. The analyses were completed separately for corn and for beans and for three different objectives. The first objective was the design of a system to achieve optimal monitoring of the average water content in the root zone, the depth of which changed throughout the growing season. The second objective aimed to produce optimal flux monitoring at a fixed depth throughout the year, located at the deepest root zone depth, which varied by crop.

3.1. HYDRUS Inputs

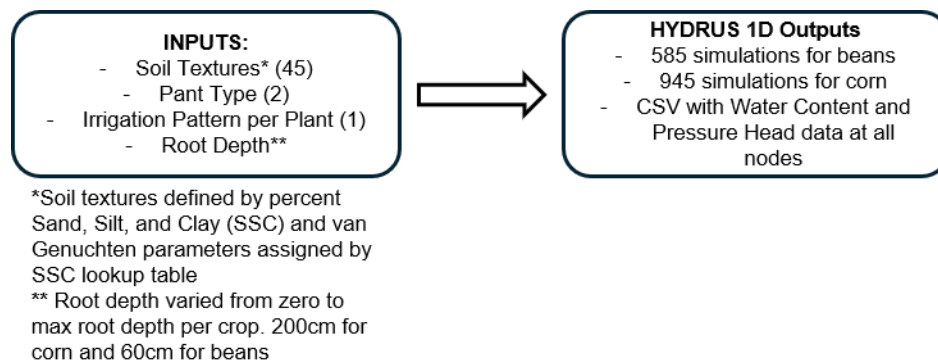


Illustration 1: Breaking the methods down into four overarching steps, the first of the five steps is illustrated here. User inputs are the foundation for all data generated by HYDRUS, which will then be passed on to train and test the decision trees.

3.1.1. Soil Selection

To ensure that the decision trees were trained on data relevant to agriculture, the soil types were limited to those in the middle of the soil texture triangle (Figure 3). Soil types were defined by percent sand, silt, and clay (SSC). No one component was allowed

to fall below 20% of the entire soil composition. Distribution of the three components were altered by 5% increments resulting in a total of 45 different soils. Soil hydraulic properties were parameterized using the van Genuchten-Mualem hydraulic model (van Genuchten, 1980; Mualem, 1976). Hydraulic parameters corresponding to each soil texture were obtained from ROSETTA, a pedotransfer database based on sand, silt, and clay fractions (Schaap et al., 2001).

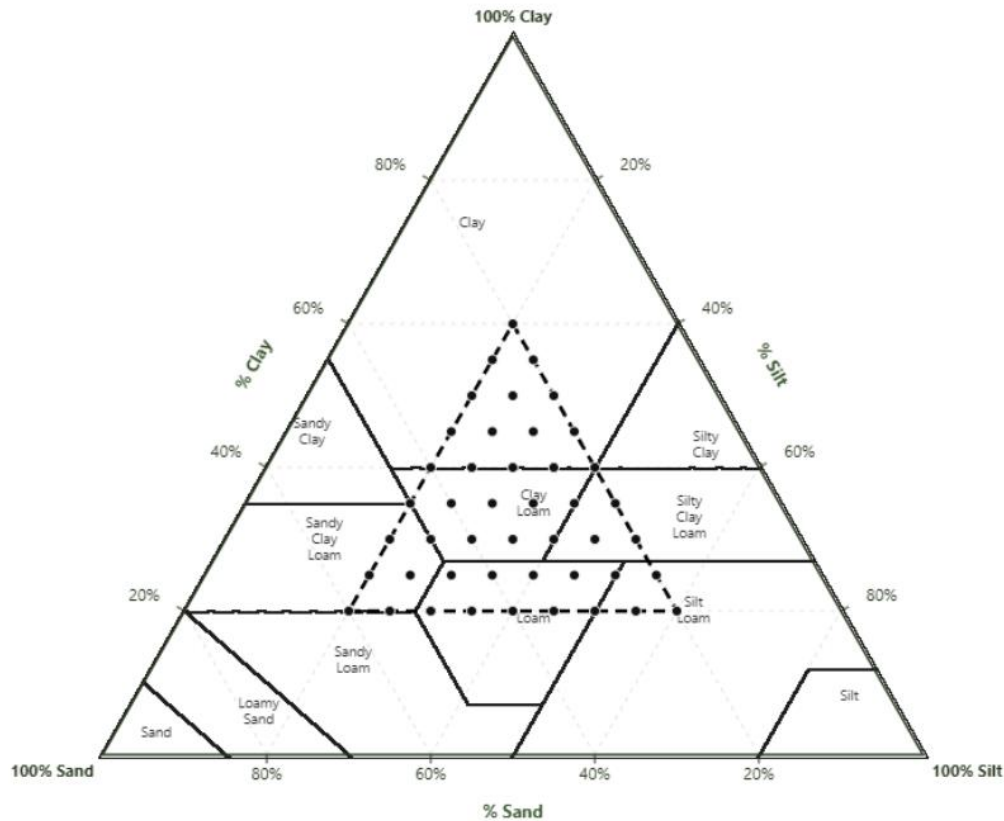


Figure 3: The USDA soil texture chart with the soil textures used in this analysis plotted in the center. By constraining the soil types not only was processing time saved, undue influence of soils that are unrepresentative of agricultural soil were excluded while still allowing the decision tree to be trained on a wide variety of soil compositions.

3.1.2. Boundary Conditions:

For this preliminary analysis, all examinations considered a 200 cm homogenous soil column with a free drainage bottom boundary condition to represent a deep water-table that does not impact near-surface conditions. Hydrus observations were available at 2 cm intervals. Flood irrigation was simulated as the top boundary condition. Flood irrigation was selected because it provides a well-defined infiltration event suitable for comparing sensor configurations under controlled conditions. To represent flood irrigation, a constant head of water (Type-I, Dirichlet boundary condition) was applied for 24 hours. The constant head value was dependent on irrigation practices of different

crops, as detailed in Table 3. The head at the top boundary was reduced to 50% of the original from hour 24 to 48. This was followed by a period of a zero-flux top boundary condition (Type-II boundary condition). The length of this type-II boundary condition was based on recommended crop irrigation cycles and is also shown in Table 3. Zero flux was chosen with the assumption that direct evaporation from the soil was minimal compared to root water uptake.

Crop	Corn	Beans
Initial Head Boundary (0-24 hours)	12cm	10cm
Secondary Head Boundary (24-48 hours)	6cm	5cm
Period of Zero Flux	15 Days	7 Days

Table 3: Duration of different boundary conditions applied to each crop. These conditions were chosen to mimic common irrigation practices for the relevant crop.

3.1.3. Initial Conditions

To represent an agricultural field that is irrigated routinely, and not allowed to drain to field capacity, initial conditions of the soil were defined based on water content. The specific initial conditions were dependent on the soil parameters. Preliminary simulations indicated that model predictions were relatively insensitive to the precise initial water content, provided the soil profile was initialized well above residual water content. Consequently, all simulations began at a uniform water-content profile equal to the midpoint between residual and saturated water content. This resulted in some internal drainage during early time, but it had little to no impact on sensor selection.

3.1.4. Root effects

A major factor that was not incorporated into Groenendyk's (2025) work on cost weighting sensor selection, but is extremely important to this analysis, is the effects of roots. To be effective and efficient, a monitoring network must maintain accuracy at all phases of crop growth. To help ensure this was considered, root depths were selected to span the full developmental range of each crop, thereby allowing sensor layouts to be evaluated throughout the growing season rather than for a single growth stage. For each crop and each soil type, root depth started at 0 cm and a new HYDRUS dataset was created for every 10 cm increment until reaching max root depth. Max root depth was set to 200 cm for corn and 60 cm for beans, which are deeper than commonly documented for both plants but represents the maximum potential rooting depths of both plants. Because the decision trees were trained on every possible combination of soil type and root depth, the sensor selections provided by the analysis offer guidance on the best possible sensor layout regardless of exact soil type or root depth. Furthermore, by including each simulation in the training, we essentially assume that all root depth stages are of equal importance for monitoring network design.

3.2. HYDRUS Output

The overarching goal of the HYDRUS runs was to generate the requisite pressure head and water content observations and corresponding outcomes of interest (average water content in the root zone and deep percolation) needed for training and testing the decision trees. Running HYDRUS for all variations of parameters resulted in 945 independent data sets for corn (45 soils with 21 root depths) and 315 independent data sets for beans (45 soils over 7 root depths). The HYDRUS outputs contained detailed hydrogeologic data for every 2 cm of the 200 cm profile depth. Following each simulation, volumetric water content and pressure head were extracted from each dataset at 4-cm depth intervals throughout the soil profile. All extracted observations for each crop type were compiled into a single final dataset which constituted the candidate predictor variables supplied to the machine-learning models.

3.3. Decision Tree Regression

3.3.1. DT Objectives

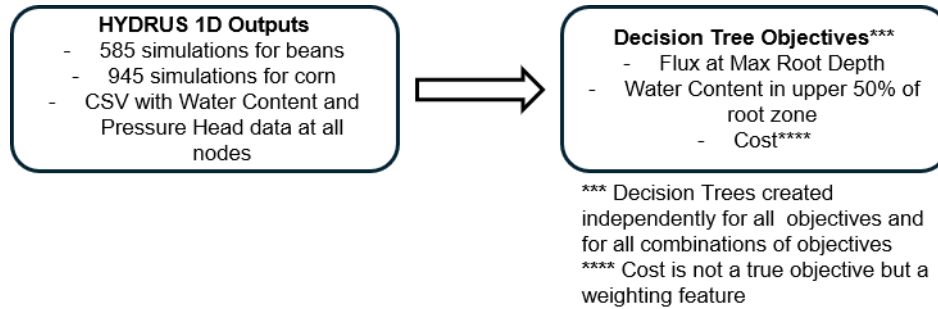


Illustration 2: The second of the five steps used in this methodology involves converting HYDRUS outputs to a data format meaningful to the decision tree objective.

Using the compiled datasets, three distinct predictive modeling objectives were considered for the decision trees, each representing a different hydrological monitoring goal under the same flooded irrigation scenario. The first objective was water flux monitoring (flux) at the deepest root zone depth over the growing season. For corn, the model predicted the water flux at a 200 cm depth. For beans flux was evaluated at 60 cm. Because flux values can span several orders of magnitude and move in either direction (downward drainage versus upward capillary rise), the target was transformed using a sign-preserving logarithmic scale before training. Therefore, the resulting error metrics should be read as indexed comparative indicators between decision trees, not actual flux values (Hastie, Tibshirani & Friedman 2009). The second objective was the average water content within the upper half of the crop's root zone (WC). The depth over which the water content was averaged was based on the specific root zone implemented for a given simulation. This target was used in its natural, untransformed form (volumetric water content in cm^3/cm^3), making its associated

error values directly and intuitively interpretable. The third and final objective was combined flux and water content prediction (Flux+WC). This model attempted to predict both above quantities simultaneously from a single shared tree structure, rather than building two separate specialized models. This tested whether a single, simpler monitoring network could serve two purposes at once, asking each sensor split in the tree to evaluate two goals rather than one. This was achieved by evaluating the reduction in combined variance. No preference was assigned to one objective over the other, only total variance reduction of the model mattered. The effect that this equal weighting had on each independent objective inside the combined model is discussed later.

3.3.2. Decision Tree Inputs

Every decision tree examined in this analysis, regardless of which prediction objective it was trained for (flux, WC, or Flux+WC model), was trained on the same underlying set of input features (pressure and water content data). These inputs represent simulated soil sensor readings drawn from the HYDRUS flood simulation. The feature set was held constant across all model variants so that differences in performance could be attributed to the modeling objective and cost-weighting strategy, rather than to differences in available input data.

Each input feature represents a single simulated sensor reading, uniquely identified by three properties: sensor type, depth (4 cm intervals), and time (daily interval). Combining these three properties (2 sensor types at 50 depths through 16-time steps) produced a total of 1,600 candidate input features made available to each decision tree during training. To evaluate whether sensors below the maximum target depth (maximum root depth) were useful, the decision trees for beans were still given access to all 1,600 input features.

3.3.3. Training and Testing

The complete dataset consisted of 945 simulated experiments. Eighty percent of the simulations (756) were randomly assigned to model training, while the remaining twenty percent (189) were reserved as an independent test dataset. Each of these trials contributed one row of input data to create the full set of 1,600 sensor readings recorded during that simulated event. These were then paired with that trial's known target values (flux, WC, or Flux+WC).

Because the candidate feature set was deliberately broad (covering the entire monitored soil profile at fine depth and daily time resolution), the decision trees in this analysis effectively functioned as sensor placement optimization tools. Each model was free to search across all 1,600 possibilities and select only those that genuinely improved prediction accuracy, subject to the cost penalty applied during training. This is what allowed for meaningful simulation of the real-world question of ‘where should a cost-limited number of sensors should be installed in a field to meet a specific objective?’ The models were not

simply tuning prediction accuracy but implicitly designing a monitoring network from a large pool of candidate locations.

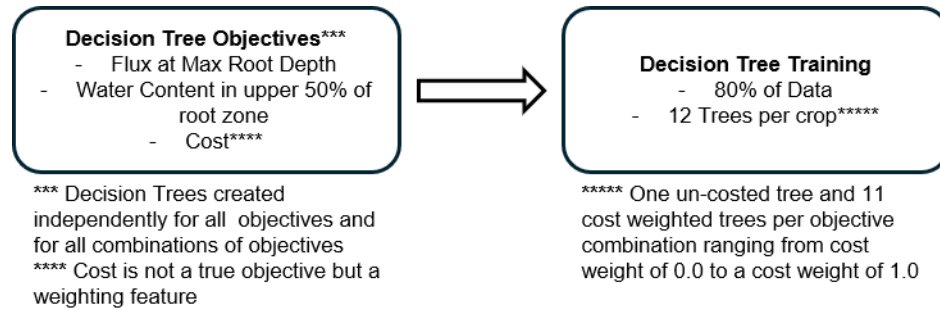


Illustration 3: With usable data from HYDRUS the decision trees can be trained on 80% of the population, generating 12 trees for each objective.

3.3.4. Cost Weighting

To compare if, and to what degree, cost weighting affected sensor layouts, the methodology introduced by Groenendyk (2025) was implemented. Twelve trees were trained and tested for each objective. A baseline tree that only focused on variance reduction (predictive accuracy), and eleven cost weighted trees where the ‘importance’ of the cost objective was set between 0.0 and 1.0 at 0.1 intervals. A threshold of 0.0 meant "weight accuracy entirely, ignore cost"; a threshold of 1.0 meant the reverse; values in between are a genuine multi-objective compromise, not a tuning knob on a single accuracy objective.

In this study, the assessed costs for sensors are not based on actual costs. Rather, they represent expected relative costs. For real-world implementation of this approach, the analysis should be repeated, including training of the decision tree, using current local costs. The sensor cost was controlled in four ways. First, an initial setup cost was assigned. This was a one-time charge applied the first time a sensor type was used anywhere in the tree and represents related monitoring equipment. A water content sensor was assigned a theoretical cost of \$4 to set up; while a pressure sensor was assigned a cost of \$10. This was done to reflect the generally higher capital costs of pressure measurement equipment relative to the monitoring equipment needed for water content probes. Once a sensor type had been introduced at any depth, this charge was not applied again. Assuming one monitoring station could monitor all probes of the same type.

Installation costs increased linearly with depth at \$0.20 per 5 cm to approximate drilling expenses and/or installation complexity. Once a sensor location existed, additional shallower sensors incurred no further installation cost. This was the only change to how cost was assigned compared to Groenendyk (2025) where cost was purely determined by sensor type and location, not other sensors used.

Sensor cost was charged once per unique sensor location (sensor type at a given depth). A water content sensor was assigned a cost of \$2.00 per installed probe; a pressure sensor was

billed at \$1.00 per unit. This was charged only the first time that specific sensor type appears at a given depth; subsequent readings from the same physical probe at different time steps were not assigned any sensor cost.

Measurement cost of \$0.10 units was charged for each individual reading (sensor type, depth, and time step combination) that appeared in the tree. This is the only recurring cost and is equal for both sensor types, reflecting the cost of taking and recording a measurement. It was charged once per unique time step at a given sensor location.

Together, these rules meant the total cost of a sensor network grew with the number of distinct sensor types used, the depths at which they were installed, and the number of time steps at which readings were taken.

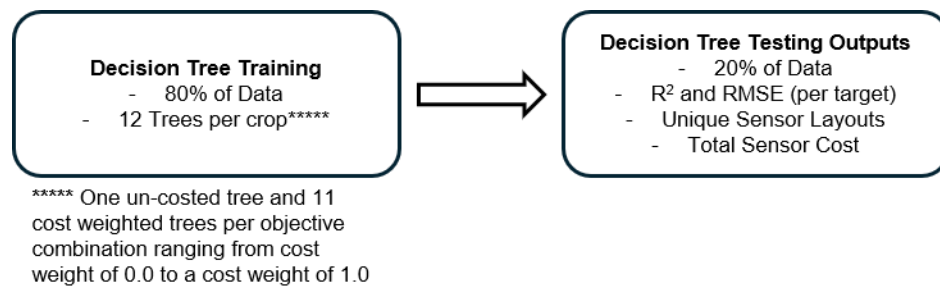


Illustration 4: The final step allows the 20% of the withheld data to be tested against the trained trees. The accuracy of the trained trees, final sensor layouts, and associated costs are provided through this testing. The testing outputs provides the foundation discussion contained in the results section.

All costs combined for all sensors used by a decision tree became the total cost of that tree. Model performance and total monitoring network cost were evaluated jointly to determine whether cost-aware decision tree regression can identify sensor configurations that achieve an improved balance between predictive accuracy and deployment cost.

4. Results

4.1 Corn

4.1.1. Overview

The initial implementation was done on corn because has large root zone variation and a larger maximum root depth, leading to more potential sensor locations. This makes corn more challenging for measurement network optimization compared to other, shallow rooting crops. This allows for more complete examination of the limits of decision tree regression for sensor

selection. All values and behaviors discussed below reference the test tree and are values for an entire irrigation cycle (15 days for corn and 7 days for beans).

Corn Overview								
Objective	Variant	Cost Threshold	R2	RMSE	Cost	Sensor Count	WC Sensors	Pressure Sensors
Flux	Cost Unaware	none	0.8958	0.001387	71.7	21	8	13
Flux	Best Cost-Aware	0.2	0.9246	0.00118	59.3	6	3	3
WC	Cost Unaware	none	0.8544	0.010193	27.2	5	2	3
WC	Best Cost-Aware	0.2	0.9382	0.006642	58.9	13	7	6
Flux+WC	Cost Unaware	none	0.7109	0.00746	65.6	19	10	9
Flux+WC	Best Cost-Aware	0	0.93	0.003767	91.4	28	13	15

Table 4: Comparison of the cost unaware and best-case cost-aware decision trees for all three objectives. Without detailed comparison of different cost weights, the comparison of ‘best to worst’ trees is misleading.

The results of sensor optimization (Table 4) are shown for no cost optimization and for limited cost optimization for the three targets considered. This initial analysis shows that accuracy is improved when some cost weighting, or at least the requirement to achieve some minimal variance reduction, is added. This may seem counter-intuitive; why should an added constraint improve variance reduction? To better understand this result, it is useful to consider the full range of cost thresholds (Table 5).

4.1.2. Flux Prediction

Corn Flux Only					
Cost Threshold	Levels	R2	RMSE	Cost	Sensors
none	5	0.8958	0.001387	71.7	21 (8wc@8-188cm, 13p@4-196cm)
0	5	0.8816	0.001479	64.7	12 (5wc@4-184cm, 7p@4-192cm)
0.1	4	0.9106	0.001285	30.4	5 (2wc@20-176cm, 3p@4-176cm)
0.2	4	0.9246	0.00118	59.3	6 (3wc@16-176cm, 3p@4-176cm)
0.3	4	0.9187	0.001225	59.3	5 (3wc@16-176cm, 2p@4-176cm)
0.4	5	0.8741	0.001525	42.6	5 (2wc@16-28cm, 3p@4-176cm)
0.5	5	0.8623	0.001595	12.4	2 (2wc@16-28cm)
0.6	5	0.8154	0.001847	8.2	1 (1wc@28cm)
0.7	6	0.8059	0.001893	8.9	1 (1wc@36cm)
0.8	5	0.8558	0.001632	16	2 (2wc@16-124cm)
0.9	8	0.4335	0.003235	11.6	1 (1wc@124cm)
1	6	0.4121	0.003295	11.2	1 (1wc@124cm)

Table 5: A direct comparison of all cost thresholds for the decision trees that predicted flux only for corn. A clear low-cost and high-accuracy zone appears around cost threshold of 0.4 to 0.5.

First, consider optimization for deep percolation (flux). When the only objective is flux prediction at 200cm the standard decision tree (no cost awareness) achieves a

respectable test R^2 of 0.896 but requires 21 sensors spanning nearly the entire depth profile (4–196 cm) at a cost of \$71.70, which is the most expensive configuration out of all the flux-focused decision trees.

Comparing this to the 0.0 cost tree (with the restriction of a minimum variance reduction) showed slightly worse performance, but with far fewer sensors and reduced cost. This suggests that the cost unaware tree may be overfitting. This is consistent with other research that has shown that some restrictions, such as adding noise to observations, can improve the performance of machine learning algorithms.

Notable improvements occurred when the trees were more heavily cost-weighted (Figure 4). At a cost threshold of 0.2, the model achieved the highest accuracy ($R^2 = 0.925$) while using only 6 sensors and facing a cost that is 17% less than the network identified with no cost consideration. To reiterate, this suggests that penalizing sensor cost during training acts as a form of regularization, helping the model avoid overfitting and arrive at a simpler, more generalized solution.

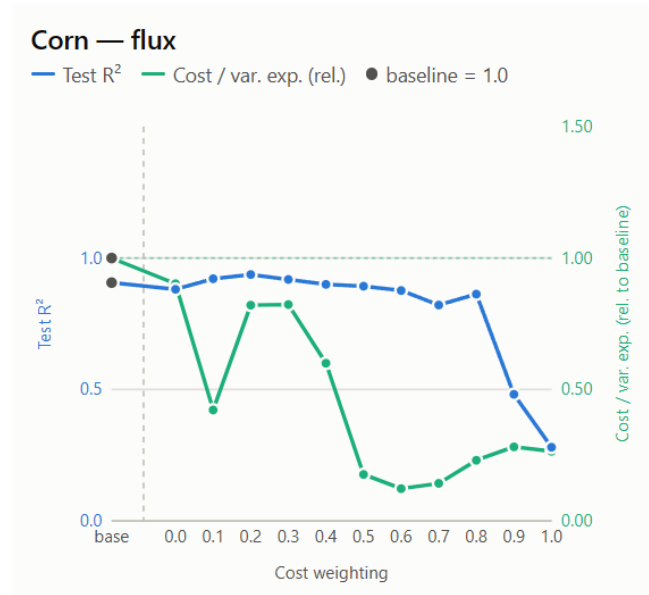


Figure 4: Comparison of R^2 values and indexed cost efficiency. Cost efficiency in this context is the total cost of the sensor layout divided by the total explained variance. This provides insight into how much one would pay per unit of improvement under that cost weighting. All cost efficiency values are indexed to the “base” or cost unaware decision tree value, which was set to 1.0. The larger the gap between cost efficiency and the R^2 value, the “better” the tree. For this objective, this gap is the greatest between a cost weight of 0.5 and 0.6.

Between cost thresholds of 0.4 and 0.6, the number of sensors included dropped from 5 to 1; the accuracy declined steadily while remaining usable (R^2 between 0.81 and 0.87). Beyond a threshold of 0.6, however, the performance decreased without seeing a corresponding reduction in cost. By a cost threshold of 0.9 and 1.0, the model collapsed to a single water content sensor at a fixed depth of 124 cm. That single sensor was used repeatedly and accuracy fell drastically to an R^2 of 0.42, which offers little practical predictive value.

In terms of RMSE, the best-performing models (cost threshold 0.1–0.3) achieve RMSE values of approximately 0.0012–0.0013. These values are the log-transformed values and carry no units, however due to the extremely low magnitude of flux at 200cm, the values are within 0.1% of the back transformed values that are in cm/day. The uncoded baseline and low-cost-penalty models sat slightly higher, around 0.0014–0.0015. As the cost penalty increased beyond 0.5, RMSE climbed steadily, reaching 0.0018–0.0019 by a cost weighting of 0.6–0.7. At the two highest cost thresholds (0.9 and 1.0), the RMSE more than doubled relative to the best-performing models (0.0032–0.0033). This sharp jump, which is consistent with the collapse of R^2 to 0.41–0.43, confirms that single-sensor models at extreme cost penalties are not simply "slightly worse," but represent a fundamentally different (and substantially less reliable) regime of monitoring network performance.

4.1.3. Water Content Prediction

Corn WC Only					
Cost Threshold	Levels	R2	RMSE	Cost	Sensors
none	4	0.8544	0.010193	27.2	5 (2wc@24-32cm, 3p@8-104cm)
0	4	0.9125	0.007904	58.9	12 (8wc@8-64cm, 4p@8-156cm)
0.1	4	0.931	0.007018	35.7	6 (3wc@4-124cm, 3p@124-160cm)
0.2	5	0.9382	0.006642	58.9	13 (7wc@4-124cm, 6p@4-176cm)
0.3	4	0.9343	0.006849	16.1	3 (3wc@16-124cm)
0.4	4	0.9336	0.006883	14.1	2 (2wc@16-124cm)
0.5	4	0.9336	0.006883	14.1	2 (2wc@16-124cm)
0.6	4	0.8477	0.010426	15.6	2 (2wc@16-124cm)
0.7	4	0.8362	0.010812	11.5	1 (1wc@124cm)
0.8	4	0.8348	0.010857	11.5	1 (1wc@124cm)
0.9	4	0.8313	0.010974	11.2	1 (1wc@124cm)
1	4	0.7606	0.013072	11.2	1 (1wc@124cm)

Table 6: Comparison of decision trees at different cost thresholds when predicting only water content for corn. Similar patterns are seen as cost-weight increases.

Similar patterns appeared when the optimization objective was the ability to characterize the season-long prediction of water content in the upper half of the root zone (Table 6). The baseline with no consideration of cost performed poorly. The performance of the test set showed an R^2 of only 0.854 while requiring 5 sensors (a mix of water content and pressure sensors spanning 8–104 cm depth) at a cost of \$27.20. While this was considerably less expensive than monitoring for flux, further reduction in cost with increased performance was achieved with some cost weighting. This again indicates that the tree that did not consider cost led to overfitting.

When optimizing for water content, there was immediate improvement when cost-weighting was introduced. However, the improvement from a small amount of cost weighting was not as clear as for flux monitoring. Specifically, as the cost weighting increased from 0.0 to 0.2, the number of sensors included decreased from 12 sensors at 0.0, to 6 at 0.1, and then rose again to 13 at 0.2. This rise in sensor count despite higher cost weighting highlights the fact that a reduction in cost and sensor count was a typical result of cost weighting, but not a hard coded objective for the decision tree. Despite the varied sensor count, the performance, as measured by R^2 or RMSE, increased steadily to a weighting of 0.2. This suggests that the cost penalty acts as a regularizing influence, reducing overfitting. Above a cost weight threshold of 0.3, the cost and performance both decrease, likely resulting in unacceptably poor performance.

A clear "sweet spot" emerged at cost thresholds of 0.3–0.5 (Figure 5), where the model became simultaneously cheaper, simpler, and more accurate than the baseline that held no consideration of cost. At a cost threshold of 0.4 or 0.5, the model achieves $R^2 = 0.934$, substantially higher than the baseline's 0.854, while requiring only 2 sensors (both water content sensors, at 16 cm and 124 cm depth) at a cost of \$14.10, roughly half the baseline cost. This is the strongest result in the table from a practical deployment standpoint: better accuracy, fewer sensors, lower cost, all at once. In practical application, the cost may be much less than indicated. Because only water content sensors were selected at a max depth of four feet, the recommended readings could be collected through direct sampling and oven drying. This would remove all sensor costs, only incurring an undetermined, but likely marginal, time cost.

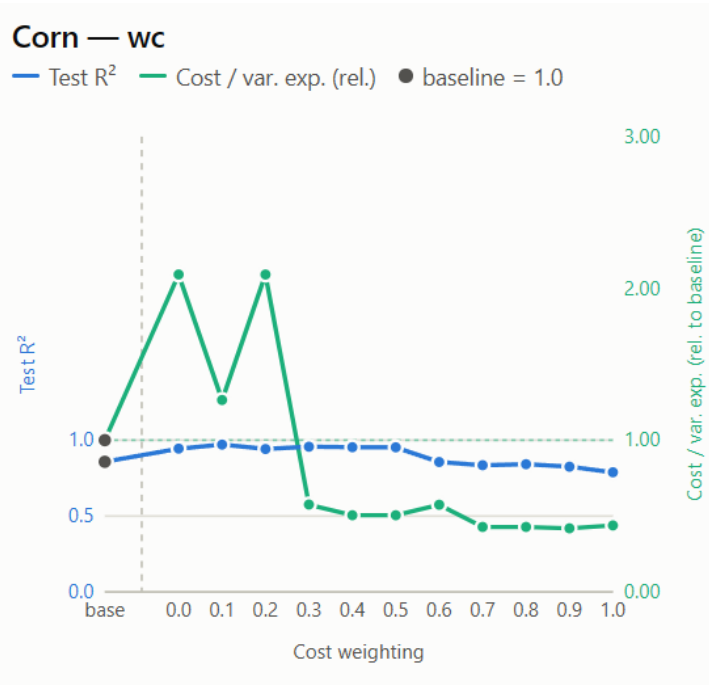


Figure 5: The comparison of cost efficiency and R^2 for WC prediction for corn across the sweep of decision trees. From a cost weight of 0.3 to 0.5 cost efficiency is optimal without R^2 dropping from previous, less cost-efficient trees.

When optimizing for the water content in the root zone, the outcome of interest was not transformed. As a result, the RMSE values in this table can be read directly as an error in the average water content, in units of volumetric water content [cm^3/cm^3]. For the best-performing configurations (cost threshold 0.1–0.5), the RMSE ranges from 0.0066 to 0.0070, meaning the model's predictions of average root-zone water content are typically within about 0.7 percentage points of the true value, a high degree of precision for a soil moisture estimate. In fact, this is a smaller error than is typically attributed to time domain reflectometry (TDR), which is considered to be the most accurate method to measure water content (Topp, G. C., Davis, J. L., & Annan, A.P., 1980; Robinson et al., 2008). The uncoded baseline performed noticeably worse with an RMSE of 0.0102. This error of more than 1% reinforces the earlier conclusion that the baseline model is not just more expensive but measurably less accurate. However, it should be noted that this is still less than a high-quality field measurements of volumetric water content obtained using calibrated TDR sensors which achieved a content measurement error of $0.02 \text{ cm}^3/\text{cm}^3$ (Robinson et al., 2008). Beyond a cost threshold of 0.6, the RMSE climbed steadily, eventually nearly doubling the error of the best-performing models. This almost two-fold increase in typical prediction error mirrors the R^2 decline observed over the same range.

4.1.4. Combined Flux and Water Content Predictions

Corn Flux+WC Content								
Cost Threshold	Combined R2	Flux R2	WC R2	R2 Gap (flux-wc)	Flux RMSE	WC RMSE	Cost	Sensors
none	0.7109	0.5665	0.8553	-0.2888	0.002829	0.0102	65.6	19 (10wc@4-172cm, 9p@4-112cm)
0	0.93	0.8971	0.9629	-0.0658	0.001378	0.0051	91.4	28 (13wc@4-64cm, 15p@4-164cm)
0.1	0.9108	0.8526	0.9689	-0.1163	0.00165	0.0047	64.8	22 (9wc@4-124cm, 13p@4-140cm)
0.2	0.8822	0.8085	0.9559	-0.1474	0.00188	0.0056	79.5	17 (7wc@4-124cm, 10p@4-176cm)
0.3	0.8942	0.8365	0.952	-0.1155	0.001738	0.0059	56.8	10 (6wc@4-124cm, 4p@4-100cm)
0.4	0.8763	0.7942	0.9584	-0.1642	0.001949	0.0054	14.5	2 (2wc@16-124cm)
0.5	0.8748	0.7903	0.9594	-0.1691	0.001968	0.0054	14.6	2 (2wc@16-124cm)
0.6	0.8762	0.7941	0.9582	-0.1641	0.00195	0.0055	14.8	2 (2wc@16-124cm)
0.7	0.8242	0.8033	0.8452	-0.0419	0.001906	0.0105	14.3	2 (2wc@16-124cm)
0.8	0.8242	0.8027	0.8456	-0.0429	0.001909	0.0105	16.3	2 (2wc@16-124cm)
0.9	0.7053	0.568	0.8426	-0.2746	0.002825	0.0106	40.1	4 (3wc@36-124cm, 1p@56cm)
1	0.61	0.4323	0.7878	-0.3555	0.003238	0.0123	42	4 (3wc@36-124cm, 1p@56cm)

Table 7: Comparison of different cost thresholds applied to the Flux+WC objective. This objective heavily favors accurate water content prediction compared to flux prediction. Of special note: water content predictive accuracy is improved in this combined objective compared to the decision tree that only aimed to predict water content.

The uncosted baseline decision tree also performed the worst when a monitoring network was designed to predict both flux and water content (Table 7). For this analysis, achieving variance reduction in either objective was valued equally as only the total variance reduction was considered. However, if a user valued the prediction of one value more than the other, this could be tuned. This would be done similarly to how cost weighting was done, by assigning scores to sensors that changed based on which objective was improved. In this case the Flux+WC objective achieved a combined test R^2 of only 0.711, driven primarily by poor ability to predict flux ($R^2 = 0.567$) despite reasonable performance for predicting water-content ($R^2 = 0.855$). This illustrates that cost-unaware multi-target trees could be problematic in general: the standard splitting algorithm, with no built-in skepticism toward weak or noisy splits, struggles even more when forced to balance two competing objectives, in this case the result is a model that uses 19 sensors yet still underperforms. Indicating that regardless of a user's actual interest in cost weighting, a cost-weight value of 0.0 produces better results than no cost weighting at all due to the limit on how fine of improvement a node may make to a systems variance.

Adding even a small cost penalty, and the associated limit on variance improvement, produced an even more dramatic accuracy improvement than for the water-content only optimization. At a cost threshold of 0.0, the R^2 jumped to 0.930, the best result of all the multi-objective trees. This occurred even though this configuration used more sensors and a higher total cost than the uncosted baseline.

While the flux-only and water-content-only analyses showed that any cost weighting resulted in lower cost and better accuracy than the baseline, that was not the case when the objectives were combined. Here, the benefit to predictive accuracy of the cost-aware training mechanism outweighed the cost difference. This caused costs to initially rise under minor cost weighting to achieve improved predictive accuracy. Cost weighting had to reach a threshold of 0.4 before the recommended sensor layout became meaningfully cheaper (Figure 6). Even when costs began to fall, accuracy was still well above cost unaware tree.

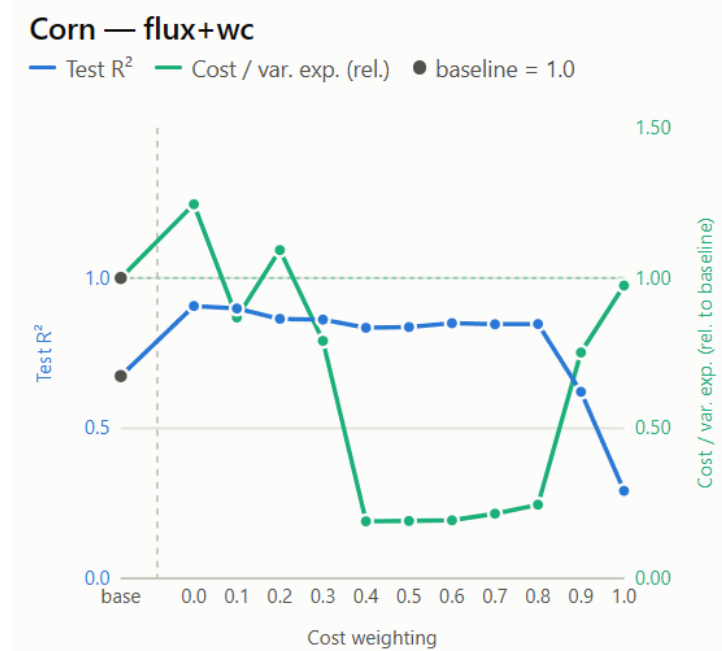


Figure 6: R^2 and cost efficiency comparison for corn's Flux+WC decision trees. The margin between R^2 and cost efficiency is the greatest at a cost weight of 0.4 to 0.8. This is a wider range and a clearer gap than with the single objective trees.

The average water content in the root zone is directly related to observed water contents. In contrast, the flux is only related to the measured water content and/or pressure head through highly nonlinear soil characteristic curves and is further complicated by the impacts of transient conditions. Therefore, it is unsurprising that the water content was consistently and substantially easier to predict than flux. This also held true for the combined water content and flux monitoring network design; the water-content predictions were always more trustworthy than the flux predictions. As implemented here, this means that if accurate flux prediction is the priority changes must be made. Either a dedicated flux-only analysis should be used over a combined analysis, or independent objective weighting should be introduced to the combined model as discussed above.

Like the single objective trees, a practical cost-efficiency "sweet spot" did emerge when designing a network to assess both water content and flux. In this case, the sweet spot occurred at a cost threshold between 0.4 and 0.6, where the model collapsed to just 2 sensors. The same two water content sensors, at 16 cm and 124 cm depth, identified as optimal in the water-content-only analysis were selected. The total cost dropped to about one-sixth of the cost of the 0.0 cost weighing configuration. Even at the lower cost, the model still achieved a respectable combined R^2 of about 0.876 (water content $R^2 \approx 0.958$ – 0.959 , flux $R^2 \approx 0.79$). This represents the best balance of cost and joint accuracy in the table.

Surprisingly, at several cost thresholds (most clearly at 0.0 and 0.1), the water-content-specific error in this Flux+WC model (RMSE WC = 0.0051 and 0.0047,

respectively) was better than the equivalent result from the dedicated water-content-only model at the same cost threshold (0.0079 and 0.0070, from the water-content table). This indicates that joint training on both targets can, in some configurations, produce synergistic benefits. The flux-related splits incidentally help refine the water-content prediction, too. The added variance reduction in flux allows sensors to be “worth the cost” compared to when the sensors were only being evaluated for WC variance reduction.

For the multi-objective trees, the flux RMSE ranged from a best case of 0.0014 (at cost threshold 0.0) to a worst case of 0.0032 (at cost threshold 1.0), roughly a doubling of error at the extreme cost setting. While the RMSE for WC ranged from a best case of 0.0047 (cost threshold 0.1) up to 0.0306 at the most extreme cost setting (threshold 1.0), more than a six-fold increase in typical error. This is a substantially steeper degradation than was observed for the same physical quantity in the water-content-only model (which topped out at 0.0131). This suggests that forcing the model to also account for flux makes its water-content predictions more fragile under heavy cost pressure than they would be in a dedicated single-purpose model.

4.1.5. Sensor Frequency and Importance

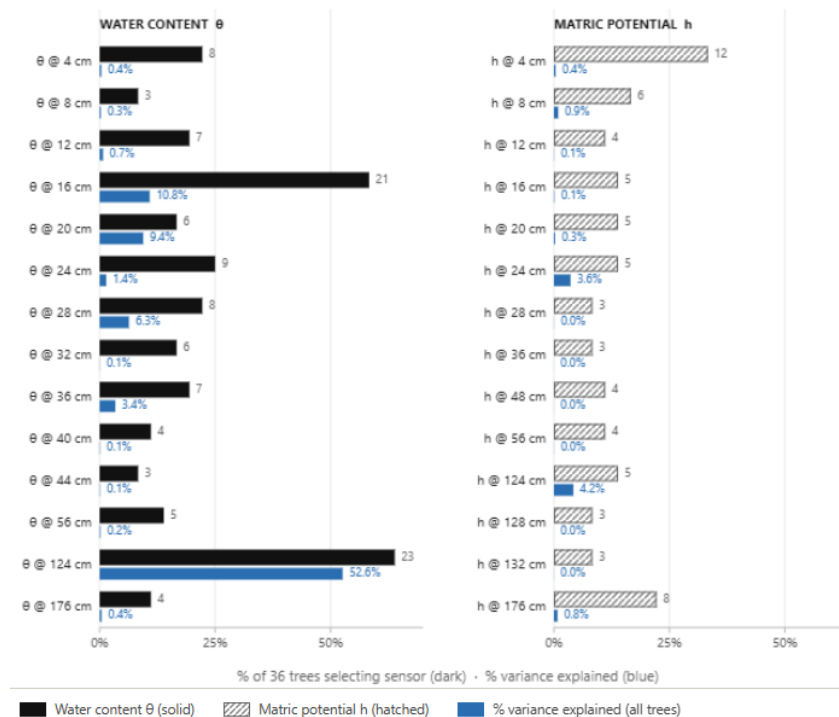


Figure 7: The number of times different sensors were chosen between all corn decision trees. Water content sensors are heavily favored and due to the deep maximum root depth, a deep-water content sensor is critical. Chosen for all the best performing trees.

While sensor selection was heavily influenced by the assigned cost weighting of the sensor types, it is worth noting that of all 36 corn decision trees the 124 cm water content sensor was chosen 23 times (64%), the 16 cm water content sensor chosen 21 times (58%), and the 4 cm pressure sensor chosen only 12 times (33%). This shows that the deep profile of corn heavily favors water content sensors and deep readings typically have some value for predictive purposes (Figure 7). However, even though sensors are frequently used due to their ability to provide some insight, that insight is sometimes marginal. For example, the 16 cm water content sensor was used by more than half of the decision trees, but across all trees combined only explained 10.2% of all variance reduction. Quantity is not inherently tied to quality in this regression model.

4.2. Beans

Beans represent a more limited case due to the shallower range of root depths but may be more representative of the complete range of agricultural crops as few other crops display max root depths as deep as corn. Analysis of beans offers an opportunity to test the broader applicability of the cost-aware decision tree approach.

4.2.1. Flux Prediction

Beans Flux Only					
Cost Threshold	Levels	R ²	RMSE	Cost	Sensors
none	3	0.8638	0.007585	32.9	7 (3wc@24-56cm, 4p@12-68cm)
0	3	0.8371	0.008295	28.7	6 (3wc@20-36cm, 3p@12-36cm)
0.1	3	0.7742	0.009767	26.4	5 (3wc@36-48cm, 2p@20-48cm)
0.2	3	0.86	0.00769	20.1	3 (1wc@36cm, 2p@20-36cm)
0.3	3	0.86	0.00769	20.1	3 (1wc@36cm, 2p@20-36cm)
0.4	3	0.804	0.009099	21.5	4 (1wc@20cm, 3p@20-44cm)
0.5	3	0.804	0.009099	21.5	4 (1wc@20cm, 3p@20-44cm)
0.6	3	0.8016	0.009155	15.4	3 (3p@20-44cm)
0.7	4	0.7916	0.009383	13.4	1 (1p@44cm)
0.8	4	0.7916	0.009383	13.4	1 (1p@44cm)
0.9	4	0.8762	0.007231	13.3	1 (1p@44cm)
1	4	-0.7971	0.027554	13.5	1 (1p@60cm)

Table 8: Comparison of cost and accuracy of different flux focused decision trees for beans. Accuracy of the cost-unaware tree is higher than the 0.0 cost threshold tree, a stark difference from the corn decision trees that were focused on flux only.

The cost unaware regression trees designed to estimate flux in beans (Table 8) achieved an R^2 of 0.864, compared to 0.896 for corn. While this is a moderate reduction, the pattern of how accuracy changed with cost weighting differed substantially between the two crops. For corn, there was an improvement in R^2 between cost thresholds of 0.1 and 0.3; test accuracy was higher than the baseline, reaching $R^2 = 0.925$. When looking at R^2 alone, no equivalent R^2

increase occurred for beans: the best cost-penalized result ($R^2 = 0.876$ at a cost threshold 0.9) represented only a marginal improvement over the baseline and was achieved at an extreme cost setting. This indicates that, for beans, the cost-penalization mechanism does not provide the same regularization benefit as for corn; the baseline model is already reasonably well-fitted without it. However, when looking at R^2 compared to cost efficiency a sweet spot does appear (Figure 8). While corn achieved the lowest cost per unit variance reduction at moderate cost weights around 0.5, beans achieved the best relationship at higher cost weights from 0.6 to 0.9.

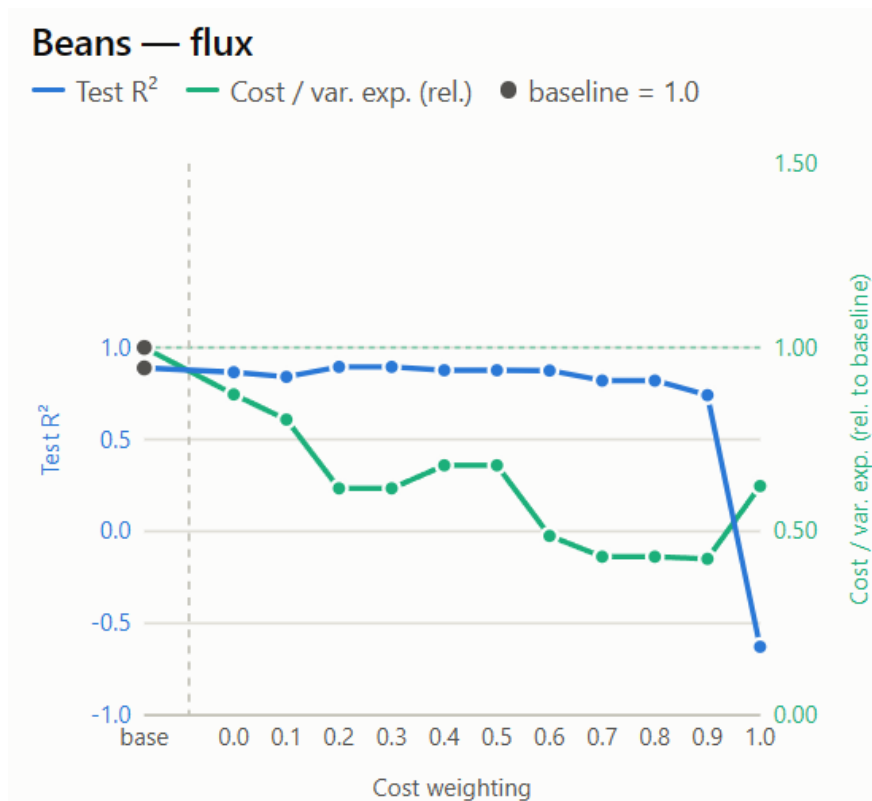


Figure 8: Comparison of cost efficiency and R^2 for the flux focused decision trees for beans. A wider gap is seen at cost weights of 0.6 to 0.9, notably later than observed in the corn comparison.

A more striking difference is the catastrophic failure at cost threshold 1.0. Optimization for flux estimation for beans at this setting produced a test R^2 of -0.797 , meaning the model performed worse than simply predicting the mean of the target variable.

The RMSE values for beans flux (0.0072–0.0276 cm/day) were substantially larger than for corn (0.0012–0.0033 cm/day). This difference is a result of physically different profile behaviors between the crops rather than an indication of worse modelling. Flux at 60 cm depth, within the active unsaturated zone during flooding, can be considerably larger in magnitude than the deep drainage reaching 200 cm in corn. For corn, most of the applied water has already been redistributed across the profile or taken up by roots before reaching the bottom of the root zone.

For beans, the water content is still more variable with depth and less of the applied water is taken up by the roots, resulting in more deep percolation. Larger flux magnitudes produced larger absolute prediction errors even when the model accounted for a similar proportion of the variance. Because RMSE figures relate to actual flux magnitudes, they therefore should not be compared directly as a like-for-like accuracy metric between crops. An RMSE of 0.001 may represent a 10% error for corn flux at 200cm but only a 1% error for bean flux at 60cm. So, if both crops were to have identical RMSE values for flux, this does not indicate equal predictive accuracy. For accuracy comparison, the R^2 values are the more appropriate cross-crop comparison.

The most notable difference between crops is in which sensor type the decision trees preferred. Decision trees that are designed to characterize flux in corn, converged strongly toward water content sensors, eliminating pressure sensors entirely. Trees aimed at characterizing flux in beans maintained a mix of water content and pressure sensors throughout the cost sweep, and at high cost thresholds (0.7–0.9) converged to a single pressure sensor at 44 cm depth. It is unclear why this result occurred, but it warrants further investigation. It is possible that, for beans, the water content at the base of the root zone (60cm), when there was significant deep percolation, correlated to significant pressure head gradients. This would increase the predictive value of pressure head readings and lead the decision trees to prefer the more sensitive sensor (pressure head) despite the added cost of using a mixture of sensor types. These trends are mirrored in the decision trees that were optimized to predict WC only.

4.2.2. Water Content Predictions

Beans WC Only					
Cost Threshold	Levels	R2	RMSE	Cost	Sensors
none	4	0.9162	0.008632	40.9	12 (7wc@4-60cm, 5p@20-64cm)
0	4	0.9172	0.008582	37.8	9 (6wc@4-60cm, 3p@20-64cm)
0.1	4	0.9083	0.009028	42.6	10 (4wc@4-24cm, 6p@4-60cm)
0.2	4	0.907	0.009093	39.5	7 (3wc@4-28cm, 4p@4-60cm)
0.3	5	0.9222	0.008318	36.1	8 (3wc@4-24cm, 5p@4-60cm)
0.4	5	0.8977	0.009539	30.7	6 (3wc@4-24cm, 3p@12-60cm)
0.5	4	0.8854	0.010095	16.1	3 (3p@12-60cm)
0.6	4	0.8844	0.01014	15	2 (2p@16-60cm)
0.7	4	0.8844	0.01014	15	2 (2p@16-60cm)
0.8	4	0.8762	0.010491	14.9	2 (2p@16-60cm)
0.9	4	0.8718	0.010676	14.8	2 (2p@16-60cm)
1	3	0.6894	0.016619	14.6	2 (2p@16-60cm)

Table 9: Comparison of cost thresholds for water content focused decision trees. Like cost-unaware flux-only tree, the cost-unaware water content tree had usable accuracy. Likely due to the shallow depth of the bean profile.

Across the water content prediction objective (Table 9), the beans and corn decision trees achieve comparable fit quality but differ substantially in the structure and robustness of the

sensor sets they rely on. Both crops' no-cost baseline trees reach their explained-variance peak at tree depth level 4, with beans achieving a marginally higher peak $R^2 = 0.910$ than corn ($R^2 = 0.907$). Test-set R^2 is similarly close at baseline (beans: 0.940; corn: 0.858). The clearest qualitative difference between the two crops is in how information is distributed across sensors. Unlike corn's reliance on a few distinct sensors, the beans water content tree draws on 12 distinct physical sensor locations spanning a range of depths, with no single sensor dominating the tree's structure. This further indicates that the beans model spreads its predictive power across a broader set of locations.

Additionally, beans response to cost weighting is less noisy, following a monotonic pattern not seen in any of the corn decision trees. This represents the more expected behavior of cost weighted decision trees and indicates that for shallow rooting plants, the more you spend the more accurate your predictions will be.



Figure 9: Comparison of the cost efficiency and R^2 for water content predictions for beans. Like with beans' flux focused trees, the best cost efficiency is seen at higher cost weight than for corn. This indicates that it may be generally 'safer' to aggressively minimize costs for shallow rooted plants than for deep rooted plants.

Further supporting the strength of the proposed approach is the ability to identify the most efficient cost weights (Figure 9). Even though R^2 remains high throughout the sweep of cost weights, the cost efficiency greatly improves at a cost weight of 0.5. Firstly, this helps inform a user who is highly conscientious of cost, just like the corn comparison. Secondly, this indicates that, compared to deep rooting crops like corn, shallow rooting crops like beans stand to have the greatest relative cost saving through cost weighted decision tree analysis.

4.2.3. Combined Flux and Water Content Prediction

Beans Flux+WC								
Cost Threshold	Combined R ²	Flux R ²	WC R ²	R ² Gap (flux-wc)	Flux RMSE	WC RMSE	Cost	Sensors
none	0.844	0.8491	0.8389	0.0102	0.00798	0.01197	91.5	13 (9wc@8-200cm, 4p@20-184cm)
0	0.8475	0.8578	0.8371	0.0207	0.00775	0.01204	28.1	6 (2wc@8-32cm, 4p@20-48cm)
0.1	0.8448	0.8309	0.8586	-0.0277	0.00845	0.01121	35	4 (3wc@8-44cm, 1p@36cm)
0.2	0.8459	0.8157	0.8761	-0.0604	0.00883	0.0105	22.7	3 (1wc@36cm, 2p@36-60cm)
0.3	0.8441	0.8365	0.8518	-0.0153	0.00831	0.01148	19.1	2 (1wc@36cm, 1p@36cm)
0.4	0.796	0.7658	0.8263	-0.0605	0.00995	0.01243	20.4	3 (1wc@12cm, 2p@12-44cm)
0.5	0.8319	0.8342	0.8297	0.0045	0.00837	0.01231	13.2	1 (1p@44cm)
0.6	0.8319	0.8342	0.8297	0.0045	0.00837	0.01231	13.2	1 (1p@44cm)
0.7	0.8319	0.8342	0.8297	0.0045	0.00837	0.01231	13.2	1 (1p@44cm)
0.8	0.6732	0.4596	0.8868	-0.4272	0.01511	0.01003	15.5	2 (2p@16-60cm)
0.9	0.6565	0.4195	0.8935	-0.474	0.01566	0.00973	15.1	2 (2p@16-60cm)
1	0.2637	-0.1663	0.6936	-0.8599	0.0222	0.0165	14.6	2 (2p@16-60cm)

Table 10: Results of the Flux+WC decision trees for beans. Unlike corn the flux-WC R² gap was not always negative.

While the no-cost Flux+WC model for corn was the worst-performing configuration across the entire analysis due to overfitting, the equivalent model for beans achieved a usable R² of 0.844 using 13 sensors (Table 10). The reason for this difference may be due to the differing complexities of the optimization problems. Corn's combined model had to simultaneously predict a target at 200cm depth (a small, slow-varying signal) and a target spanning the top 100cm of the profile, using a tree that must serve both objectives at once across a large training dataset spanning many root depths. Beans' shallower profile and shorter simulation duration produced a simpler, more consistent dataset where the combined objective was less conflicted.

One of the most notable contrasts between crops appears in the per-target R² breakdown. For corn, water content was consistently and substantially better predicted than flux for every configuration. In contrast, this pattern was largely absent for beans. The reason for this difference is unclear. It may be that this reflects the larger range in fluxes seen for beans. Or it could be due to the larger water content variation within the top half of the root zone, making the average more difficult to determine from a small number of measurements. While this warrants further investigation, it is important to note that, for this study, the ability of the cost-aware decision tree to identify these different monitoring networks, independent of an underlying conceptual model, indicates the strength of the proposed approach.

Unlike corn, neither water content sensors nor deep sensors were chosen a disproportional number of times (Figure 10). The opposite occurred, with pressure sensors representing 81 of the 129 sensors selected (62%). The most chosen sensor was at 20cm, only one third of the way down the soil profile, compared to corns deepest sensor at 124cm which is placed more than 60% of the way down the total profile. Additionally, despite the decision trees being given access to the full range of sensors, the bean decision trees never selected a sensor

below 60cm. Indicating that information beyond the target depth is irrelevant. Finally, while there are sensors selected more frequently than is justified by the variance explained by those sensors, the contrast is not as stark as it was for corn.

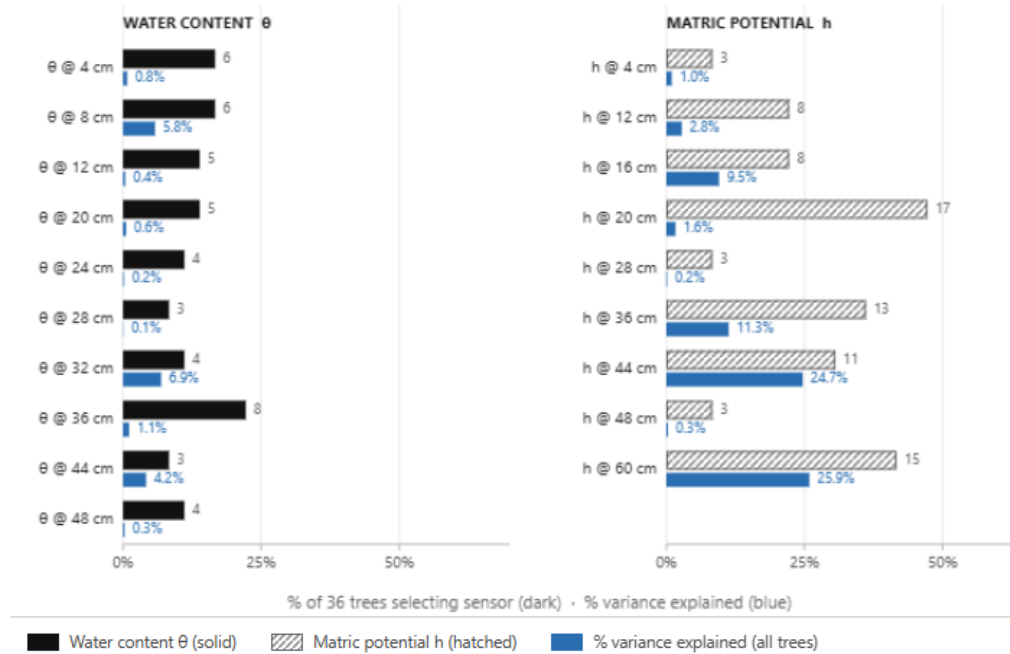


Figure 10: The number of times sensors were chosen, in total, between the 36 bean decision trees. The decision trees for beans highly favored matric potential sensors rather than water content sensors as was seen in the corn decision tree results.

Another unique feature for beans' Flux+WC decision trees was that the gap between cost efficiency and R^2 was not as localized as all other previously seen objectives (Figure 11). While the largest gap between cost efficiency and R^2 still appears, at cost weights of 0.5 to 0.7, the gap is not as uniquely large compared to the other cost weights as before. This indicates, for shallow rooting plants such as beans, there may be a wider range of sensor layouts available to a user who is focused on cost efficiency, rather than just raw cost.

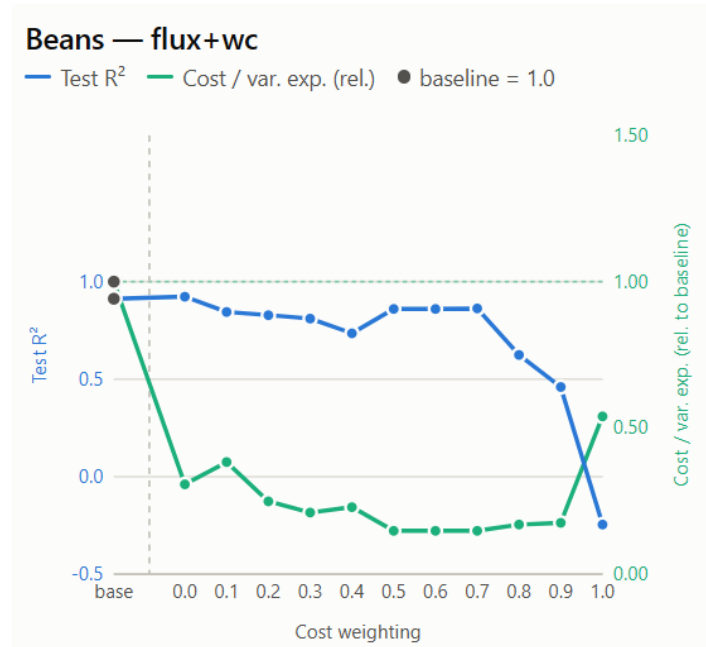


Figure 11: Cost efficiency and R^2 comparison of beans' multi-objective decision trees. While the previous analyses for corn and beans' single objective trees showed specific zone where cost efficiency was the best, it is not as clear for beans.

5. Discussion

5.1. Revisiting the Questions

As discussed in the introduction, this research aimed, first and foremost, to determine if cost weighted decision trees were a useful and practical tool to prescribe sensor layouts in any form. Secondly, assuming the first goal proved valid, this research aimed to determine whether it is possible to proscribe a universal sensor layout, or whether optimization must be soil- and crop-specific.

5.2. Overall Applicability

For the first research goal to be supported, the decision trees for any one objective must prescribe sensor layouts that are logical, insightful, and meaningfully different from current methods. Additionally, there must be measurable differences between the range of cost-weighted decision trees that allow for informed choices to be made by the user. Overall, the deployment of soil sensor networks for irrigation management presents a well-established optimization problem. It is generally assumed that the accuracy of characterization improves with the number of sensors and the number of measurement times. However, in practice, sensor installation, and maintenance costs constrain the number of observations that can be collected. Conventional approaches to sensor network design typically rely on physical intuition, rules of thumb, or iterative field trials, none of which systematically quantify the trade-off between information

gain and cost. Decision tree regression, particularly when augmented with a cost-aware splitting criterion, offers a principled, data-driven alternative that explicitly models this trade-off during the learning process itself. The results presented here, derived from HYDRUS-1D simulations for two crops under flood irrigation over a wide range of soils, provides an initial evaluation of a novel approach to monitoring network design. This approach can be modified easily to accommodate different measurement costs, crops, irrigation practices, or soil types. It also allows a user to choose the relative importance of prediction accuracy and cost. Finally, while two optimization targets are considered (average water content in the top half of the root zone and deep drainage), the method could be applied to any outcome(s) that could be produced with a model constrained by observations.

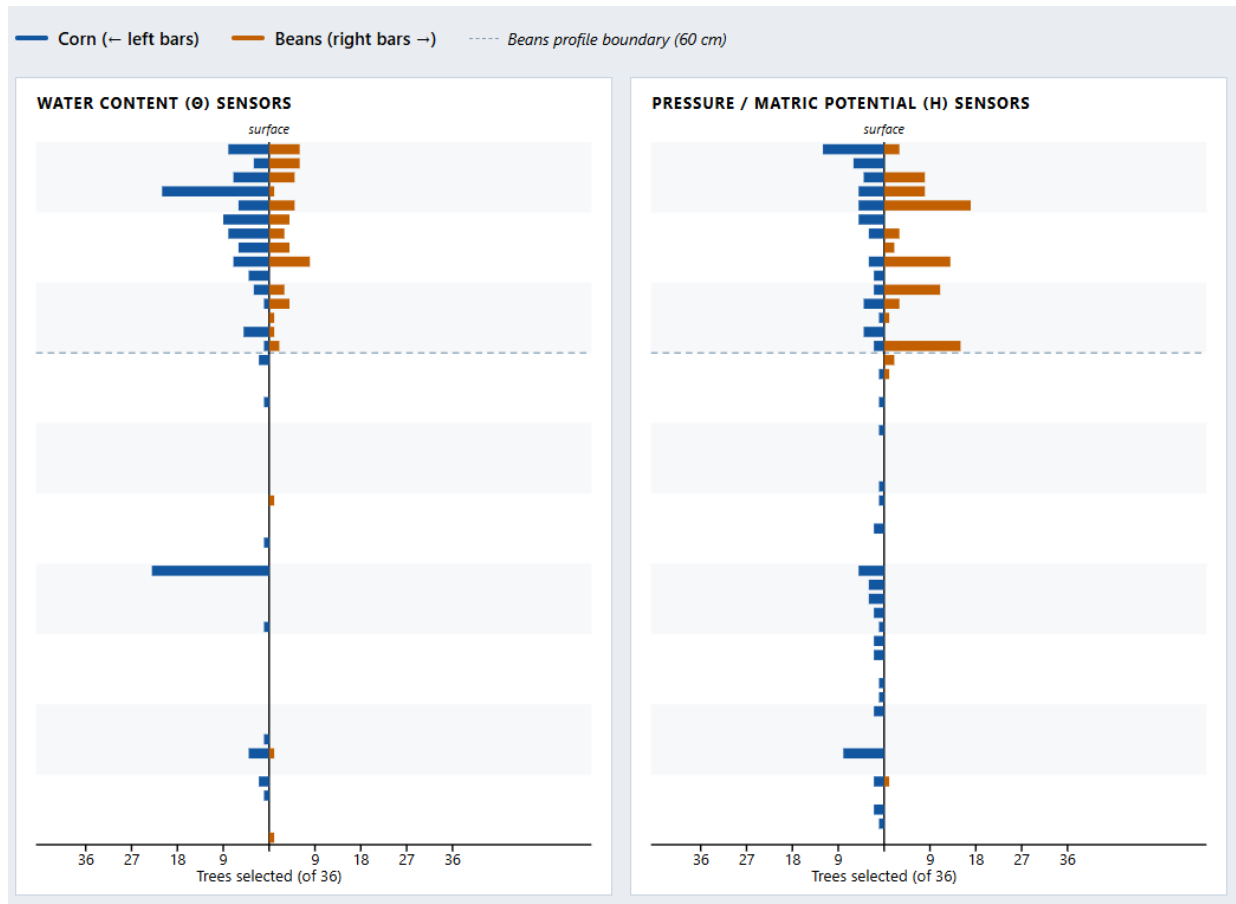


Figure 12: Direct comparison of sensor selections by depth between corn decision trees and bean decision trees. Corn is much more reliant on water content sensors and beans are more reliant on matric potential sensors. This indicates that any future regression analysis for site specific sensor layouts must be trained on data that accurately represents the profile of the crop to be grown.

Additionally, this investigation showed that most of the proposed measurement network designs were consistent with our current understanding. The types and depths of sensors were sensible, if at times surprising (Figure 12). This is important because the proposed networks confirm our basic intuition. For example, it is more difficult to predict shallow percolation that varies widely than to predict relatively invariant deeper percolation. This gives us confidence in the more surprising findings, such as that while water content sensors are preferred for predicting

average water content in the root zone, a mixture of water content and pressure sensors are selected for characterizing flux beneath beans. The findings also indicate that there is very little improvement in the accuracy of estimates of average water content or flux associated with adding more than a few well-placed sensors (Figure 13).

Finally, specific to using machine learning, the findings suggest that the regularization afforded by cost-awareness may improve sensor network design costs. Admittedly, a weakness of these high-efficiency sensor networks is that they lack robustness. Sensors, as physical components of a system, are susceptible to failure and with so few sensors present in a cost-optimized layout a single failure may collapse the entire system. Because non-cost optimized sensor layouts, with inherent redundancy, also bear failure costs this is not a point of failure for cost optimized sensor layouts. Instead, it just increased the importance of rapid replacement of failed sensors effectively trading redundancy for urgency.



Figure 13: Accuracy and cost trends for different objectives by crop type. In most cases, predictive accuracy is maintained at a higher level while costs decrease. Indicating that cost-weighting is beneficial and regression analysis stands to be a cost-saving tool for the agricultural industry.

It is important to recognize that all the analyses presented are based solely on simulated results. While it can be argued that soil physical principles, such as are the basis of HYDRUS

1D, are well established, it is also true that the simplified conditions of homogeneity, constant initial water content, and perfectly defined boundary conditions used in these analyses are not strictly realistic. On the other hand, the same physically based models and/or numerical approximations are always applied to analyzing unsaturated flow. Therefore, this analysis, despite its limitations, presents the best possible current approach to examining the optimization of monitoring networks for irrigation efficiency. Considering the above discussion, the initial question of the research is answered in the positive. Cost-weighted decision trees are, in general, a practical tool that allows users to make data driven decisions. These decision trees support a user's decisions regardless of whether they are focused on raw network costs, or network cost efficiency.

5.3. Universal vs. Site-Specific Sensor Layouts

With the first research question answered in the affirmative, the second question may be evaluated. At a scale that ignores crop type, no universal sensor layout, or even most cost-efficient sensor layout, may be identified through cost-weighted decision tree analysis. Referencing Figures 7, 10, and 12, the type and depth of sensors vary too greatly between crops for this to be possible. When the type of crop, and correlating objectives, is identified an application of a universal sensor layout is still impossible. While each crop there are highly utilized sensors, this does not translate to a universal sensor for that crop. For corn a water content sensor at 124 cm appears more frequently than any other, and for beans pressure sensors at 60 cm, 20 cm, 44 cm, and 36 cm are highly common across all the trees. However, neither corn's nor beans' most used sensors can be called 'universal.' This is because per objective, no sensor is selected 100% of the time across all trees and most are chosen by less than 50% of the trees (Table 11).

Crop	Objective	Total Unique Sensors Selected Between All Decision Trees	Selection Rate = 100%	Selection Rate >75%	Selection Rate > 50%	Selection Rate ≤ 50%
Beans	Flux	14	0	0	P @ 20 cm and 36 cm	12
Beans	WC	20	0	P @ 60 cm	p @ 20 cm	18
Beans	Flux+WC	19	0	0	0	19
Corn	Flux	26	0	0	WC @ 16 cm	25
Corn	WC	24	0	WC @ 124 cm	0	23
Corn	Flux+WC	46	0	WC @ 124 cm	WC @ 16 cm	44

Table 11: Breakdown of how often sensors were selected across a range of trees per objected by crop type. At no point was a sensor always used, most sensors were not even chosen by 50% of the trees that had the same objective but different cost-weights.

The most chosen sensor, a water content sensor at 124 cm, was selected by 83% of the decision trees that aimed to predict flux at 200 cm. This sensor was chosen less than 50% of the

time when the objective was to predict water content in the upper half of corn's root zone. All other sensors had even lower selection rates, even when the type of crop was specified. Considering this, the second research question is answered in the negative. Universal sensor layouts cannot be identified, new regression analysis should be performed for each site, for each crop at the site, and for each objective of interest.

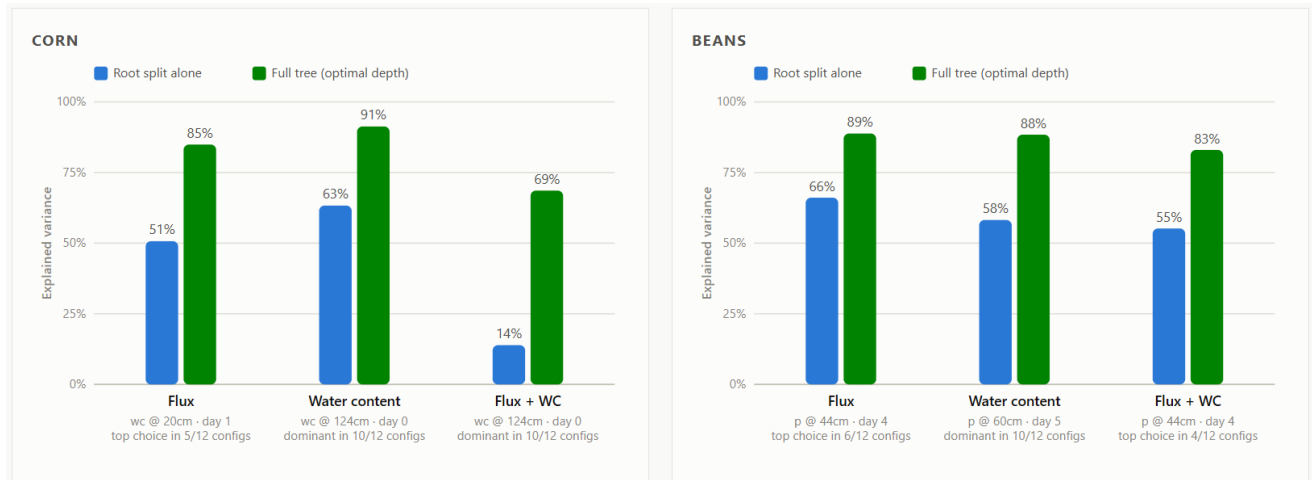


Figure 14: The sensor (across all 12 trees of an objective) that independently explains the most variance. Interestingly, for no objective is the most effective sensor that is shown in the graph present in all 12 trees.

What may be identified however, is the most important sensor per crop per objective. At least the most important sensor in general may be identified. Figure 14 shows the most effective sensor for each set of trees, but even when a single sensor explains 90% of the variance (averaged across the 12 trees) that sensor does not appear in all trees for any objective. For trees where the most effective sensor is used, it is almost always supported by other sensors.

5.4. Improvements and Future Work

The results suggest several productive directions for further development. First, validation of trained models against measured field data across a range of soil textures and crop types would establish the degree to which simulation-trained trees transfer to real conditions and would identify the scenarios in which retraining on field data is necessary. Because we cannot measure water flux directly in the field, this may be better suited to assessing salinity issues. The interaction between measurement costs and network design could lead to important insights into which measurement types should be improved and at what cost. Or this could be used to re-examine the possible utility of simpler, less accurate measurements such as sample collection with oven drying. Some of the user-selected controls such as the cost threshold, measure of improvement (e.g. variance reduction), and weighting of multiple objectives could be improved. Consideration of multiple realizations to ensure optimal tree selection and, therefore, optimal network design could be implemented. The analysis could be extended to different irrigation

scenarios, more plants, and, potentially, commonly occurring layered profiles. Additionally, the tool could be made more user-friendly and combined with decision support tools to allow farmers to use it to design purpose-built monitoring networks and to improve their understanding of the value of monitoring for their irrigation practices.

Timing of Sensor Readings for Corn						
Objective	Number of Trees	Number of Unique Sensors	Early Cycle Readings (day 0-4)	Mid Cycle Readings (day 5-9)	Late Cycle Readings (day 10-15)	Total Readings
Flux	12	78	63	19	53	135
Water Content	12	62	50	16	26	92
Flux+WC	12	186	142	67	81	290
TOTAL	36	244	255	102	160	517

Table 12: The timing of all readings for Corn's different objectives is shown above. The accuracy of Corn's decision trees seems to rely less on mid-cycle readings of sensors and draw greater data from early and late-cycle readings.

Timing of Sensor Readings for Beans						
Objective	Number of Trees	Number of Unique Sensors	Early Cycle Readings (day 0-2)	Mid Cycle Readings (day 3-5)	Late Cycle Readings (day 6-7)	Total Readings
Flux	12	26	27	30	10	67
Water Content	12	46	52	43	15	110
Flux+WC	12	46	44	27	6	77
TOTAL	36	94	123	100	31	254

Table 13: The timing of sensor readings for beans indicates that late cycle readings are less important for the shallower rooting crop.

Finally, testing different irrigation periods representative of actual irrigation practices may identify critical measurement periods. Tables 12 and 13 show generally when readings were taken by objective. For corn mid cycle readings seem less important than early and late cycle reading. Beans, however, seem to be better served by mid-cycle reading with late cycle readings being chosen less often. This needs to be investigated in greater detail to draw actual conclusions. How the readings were binned was arbitrary, and the chosen length of the irrigation cycles is likely to heavily influence the results. So, while there is indication that decision tree regression also provides insight to the best window for observations, conclusions on that matter may not be drawn from the analysis done above.

6. Conclusion

The results presented here indicate that cost-aware decision tree regression is a technically viable and physically interpretable method for identifying cost-optimized soil sensor configurations for irrigation monitoring. The tool consistently identified sensor networks that improve accuracy relative to decision trees that do not consider cost while substantially reducing the number of sensors used. This reduction in sensor count returned up to 80% savings with negligible change to predictive accuracy. Implementation of this method stands to be an easy yet powerful tool for achieving cost reduction for farmers. These findings were replicated for two

crops and three prediction objectives (average water content in the root zone over a growing season, percolation beneath the maximum root depth, and the combination of the two). These promising initial results suggest that it is worthwhile testing cost-aware decision trees further for the design of monitoring networks for irrigation efficiency or for related hydrologic processes.

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